

ANURAG KUMAR SINGH

MLOps & System Design Engineer

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SUMMARY

MLOps and System Design focused engineer with hands-on experience building scalable ML deployment pipelines, real-time inference systems, and distributed infrastructure. Google internship experience in cloud ML infrastructure, backed by a strong competitive programming foundation (ICPC Regional Finalist, Codeforces Master). Focused on bridging the gap between ML research and reliable, production-grade systems.

EXPERIENCE

Software Engineering Intern — Google

May 2025 – Jul 2025

Cloud AI/ML Infrastructure Team, Bengaluru, India

- Designed and shipped an automated ML model retraining & deployment pipeline using Kubeflow and Vertex AI, cutting model refresh turnaround from 5 days to under 8 hours.
- Built a real-time model monitoring and drift-detection service (Python, Prometheus, Grafana) covering 12+ production models, reducing incident response time by 40%.
- Redesigned the internal feature store's caching layer with Redis and gRPC, improving feature-lookup latency by 35% at 50K+ requests/minute peak load.
- Collaborated with a 6-person cross-functional team of engineers and researchers; presented the pipeline design at a system design review attended by senior staff engineers.

PROJECTS

ServeMesh — Distributed Model Serving Platform

Kubernetes, Istio, Docker, TorchServe, ONNX Runtime, Kafka, Prometheus, React

- Problem:** ML teams were deploying models via ad-hoc Flask servers with no autoscaling, versioning, or fault tolerance, causing frequent downtime during traffic spikes.
- Solution:** Built a self-service platform letting teams deploy any model behind a versioned, autoscaled endpoint in under 5 minutes, with automatic rollback on failure.
- Built with:** Kubernetes-based serving layer with Istio for canary rollouts and traffic splitting, TorchServe/ONNX Runtime for inference, Kafka for async request queuing, and a React dashboard for deployment management. Achieved 99.9% uptime and sub-100ms p95 latency across 20+ models in load testing.

Real-Time Fraud Detection System

XGBoost, LightGBM, Kafka Streams, Apache Flink, Redis, gRPC, Docker

- Problem:** Needed to score financial transactions for fraud risk within milliseconds, at ~10,000 transactions/sec, without blocking checkout flow.
- Solution:** Delivered sub-50ms fraud scoring via a streaming feature pipeline, cutting false negatives by 22% versus the legacy batch-scored system.
- Built with:** Gradient-boosted ensemble (XGBoost + LightGBM) with online feature computation through Kafka Streams and Flink, served via a low-latency gRPC microservice backed by a Redis feature cache; validated with shadow-traffic A/B testing before full rollout.

PipeForge — AutoML Pipeline Orchestrator

MLflow, Optuna, Apache Airflow, Docker, Python, PostgreSQL

- Problem:** Data science teams lost significant time to repetitive experiment tracking, manual hyperparameter tuning, and reproducibility issues across projects.
- Solution:** Built an end-to-end MLOps orchestration tool automating experiment tracking, hyperparameter search, and CI/CD-style model promotion, cutting average model iteration time by 60%.
- Built with:** MLflow for experiment tracking, Optuna for Bayesian hyperparameter optimization, and Airflow for pipeline orchestration, packaged as reusable Docker images with a YAML-based config so any team could plug in a new model in under 30 minutes.

ACHIEVEMENTS

- 4x Hackathon Winner — Smart India Hackathon (2026), HackHarvard (2025), ETHIndia (2025), Google Solution Challenge Regional Finals (2025)
- ICPC Regional Finalist (2024) — Ranked among the top teams at the Kanpur Regionals
- LeetCode: 500+ problems solved, Knight badge (Rating ~2050) — leetcode.com/anuragsk
- Codeforces: Master, Max Rating 2150
- CodeChef: 4★ Rated, Max Rating 1850 — 360+ problems solved across competitive programming platforms

SKILLS

Languages: Python, C++, Java, SQL, Bash

Data Science & ML: NumPy, Pandas, Scikit-learn, Statistical Modeling, Feature Engineering, Exploratory Data Analysis

Deep Learning: PyTorch, TensorFlow, Keras, CNNs, RNNs/LSTMs, Transformers, Hugging Face

MLOps: MLflow, Kubeflow, Apache Airflow, Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), Model Monitoring, Feature Stores

System Design: Distributed Systems, Microservices, Load Balancing, Caching (Redis), Message Queues (Kafka), API Design (REST/gRPC), Database Design (SQL/NoSQL)

Cloud & Tools: AWS (SageMaker, EC2, S3), GCP (Vertex AI, BigQuery), Git, Linux, Prometheus/Grafana

EDUCATION

Heritage Institute of Technology, Kolkata

2024 – 2028

B.Tech in Computer Science & Engineering — CGPA: 8.7 / 10

Relevant Coursework: Distributed Systems, Machine Learning, Operating Systems, Database Management Systems, Cloud Computing